Performance modeling for Computer systems (Midsem 40 Marks)

1. Consider an $M/M/\infty$ queue.(10mks)

   (a) Model it as a continuous-time Markov chain and write the Q matrix and transition diagram. (3mks

   (b) Show that the number of jobs in an $M/M/\infty$ queue in steady state is Poisson distributed. What is the parameter ? (7 marks)

2. Consider an arrival process which is Poisson with rate λ, and let $K \in \mathbb{N}$ be fixed. Compare the following queueing systems, all with exponential service times of rate μ per server. (You need to derive stationary distributions wherever required)

   (a) An $M/M/1/\infty$ queue with a single server of service rate $K\mu$.

   (b) An $M/M/K/\infty$ queue with K identical servers, each with service rate μ, and a single central queue.

   (c) K parallel $M/M/1/\infty$ queues, each with service rate μ, where each arriving job joins one of the K queues uniformly at random, independently of the current queue lengths.

   Assuming all systems are stable, obtain $\mathbb{E}[N]$ for all systems. ALso identify which system has the smallest mean number of jobs in steady state, $\mathbb{E}[N]$. Justify your answer.
   [10 marks]

3. State three equivalent definitions of a Poisson process. Prove equivalence between any two of them.
   [10 marks]

4. Consider a continuous-time Markov chain with generator matrix $Q = (q_{ij})$. Suppose there exists a probability vector $\boldsymbol{\pi} = (\pi_i)$ such that

$$\sum_i \pi_i = 1 \quad \text{and} \quad \pi_i q_{ij} = \pi_j q_{ji} \quad \text{for all } i \neq j.$$

   (a) Show that $\boldsymbol{\pi}$ is a stationary distribution of the Markov chain. (5mks)

   (b) State the criteria for classifying state i of a DTMC as recurrent (null and positive) and transient based on $F_{ii}, \mu_{ii}, f_{ii}^n$ and $\sum_n p_{ii}^n$.(5mks)